**1** An object of mass 0.050 kg is projected from the top of a 50 m tall building at an angle of 60° to the horizontal with an initial speed of 20 m $s^{-1}$ as shown in Fig. 1.1.

**A** is a point in the trajectory that is at the same height as the object initially and *d* is the horizontal distance the object lands from the building.

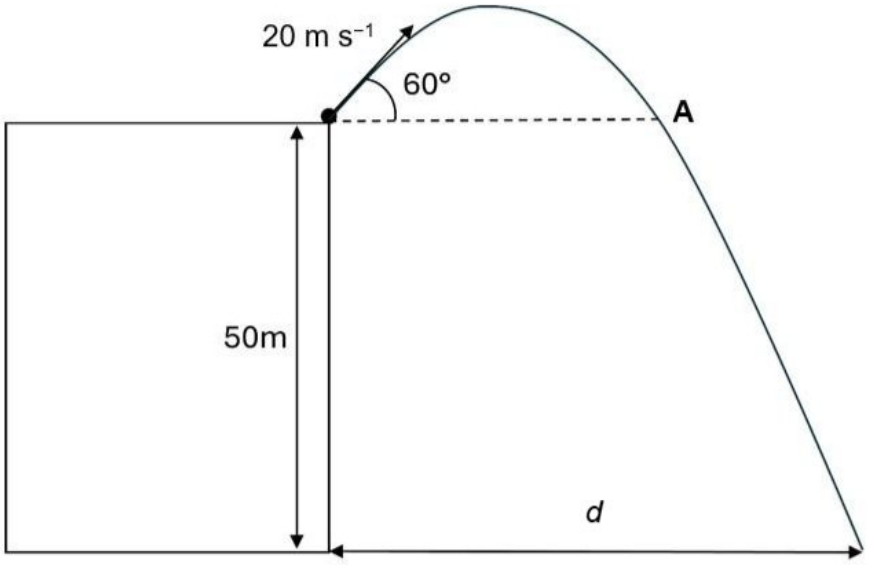



**Fig. 1.1**

**(a)** Assuming there is no resistance.

**(i)** Calculate the distance *d*.

*d* = ..... m [3]

**(ii)** Without calculation, state the speed of the object at **A** and explain how you arrived at your answer.

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..... [2]

**(b)** Experimentally, it is found that the same object now reaches a maximum height of 60.5 m above the ground in its trajectory.

**(i)** Show quantitatively that the air resistance is not negligible in this case.

[2]

**(ii)** The speed of the object at its maximum height is 8.0 m $s^{-1}$.
Calculate the energy lost due to air resistance at maximum height.

energy lost = ..... J [2]

**(c)** Sketch the path of the object in Fig. 1.1 when air resistance is not negligible. [1]

[Total: 10]